

SAFETY DATA SHEET (SDS)

This safety data sheet complies with the requirements of:
Regulation (EC) No. 1907/2006 and Regulation (EC) No. 1272/2008, (EU) No. 2015/830

Revision Date 07-Aug-2023

WAI2 - EGHS - EUROPEAN

Revision Number 4

SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product Identifier

Product Name ORP Standard

Product No 967961
Unique Formula Identifier (UFI) Not applicable

REACH registration number Not applicable

Pure substance/mixture Mixture

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Use as laboratory reagent

Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Manufacturer, Importer, Supplier Thermo Fisher Scientific
Robert-Bosch-Str. 163505
Langenselbold, GERMANY
Tel.: +49 (6184) 90-6000

E-mail address info.water@thermo.com

Made in USA

1.4. Emergency telephone number 24 Hour Emergency Phone Number
CHEMTREC®
Within USA and Canada: 1-800-424-9300
Outside USA and Canada: 1-703-527-3887
(collect calls accepted)

SECTION 2. HAZARDS IDENTIFICATION**2.1. Classification of the substance or mixture**

Classification - Mixture

Classification according to Regulation (EC) No. 1272/2008 [CLP]

Specific target organ toxicity (repeated exposure)	Category 1
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2.2. Label elements**Signal Word**

Danger

Hazard Statements

H372 - Causes damage to organs through prolonged or repeated exposure

Precautionary Statements

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P314 - Get medical advice/attention if you feel unwell

P501 - Dispose of contents/ container to an approved waste disposal plant

2.3. Other hazards

No information available

General Hazards

This product does not contain any known or suspected endocrine disruptors
Toxic to terrestrial vertebrates

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Component	EC No	CAS No	Weight %	CLP Classification - Regulation (EC) No 1272/2008	REACH Reg. No
Water	EEC No. 231-791-2	7732-18-5	50 - 60%	Not classified	No information available
Potassium Iodide	EEC No. 231-659-4	7681-11-0	40 - 50%	Acute Tox. 4 (H302) Skin Irrit. 2 (H315) Eye Irrit. 2 (H319)	No information available
Boric Acid	EEC No. 233-139-2	10043-35-3	0 - 10%	Repr. 1B (H360FD)	No information available
Iodine	EEC No. 231-442-4	7553-56-2	0 - 10%	Acute Tox. 4 (H312) Acute Tox. 4 (H332) Aquatic Acute 1 (H400) STOT RE 1 (H372)	No information available
Potassium Hydroxide	EEC No. 215-181-3	1310-58-3	0 - 10%	Acute Tox. 4 (H302) Skin Corr. 1A (H314)	No information available

Component	CAS No	Specific concentration limits (SCL's)	M-Factor	Component notes
Water	7732-18-5	-	-	-
Potassium Iodide	7681-11-0	-	-	-
Boric Acid	10043-35-3	Repr. 1B :: C>=5.5%	-	-
Iodine	7553-56-2	-	-	-
Potassium Hydroxide	1310-58-3	-	-	-

SECTION 4: FIRST AID MEASURES**4.1. Description of first aid measures**

General Advice	If symptoms persist, call a physician.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.
Inhalation	Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.
Ingestion	Clean mouth with water and drink afterwards plenty of water.
Self-Protection of the First Aider	No special precautions required.

4.2. Most important symptoms and effects, both acute and delayed

Most important symptoms and effects	None reasonably foreseeable
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4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician	Treat symptomatically
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SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable Extinguishing Media

No information available

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Personal Precautions

Ensure adequate ventilation. Use personal protective equipment as required.

6.2. Environmental precautions

Environmental Precautions

Should not be released into the environment. See Section 12 for additional Ecological Information. Vapors may accumulate to form explosive concentrations.

6.3. Methods and material for containment and cleaning up

Methods for Containment

Prevent further leakage or spillage if safe to do so.

Methods for Cleaning Up

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust).

Reference to Other Sections

Refer to protective measures listed in Sections 7 and 8

See Section 8 for information on appropriate personal protective equipment

See Section 12 for additional Ecological Information

See Section 13 for additional waste treatment information

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Advice on safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

General hygiene considerations

Handle in accordance with good industrial hygiene and safety practice.

7.2. Conditions for safe storage, including any incompatibilities

Storage Conditions

Keep container tightly closed in a dry and well-ventilated place. Store at room temperature in the original container. Protect from direct sunlight.

7.3. Specific end use(s)

Specific Use(s)

Use as laboratory reagent

Risk Management Methods (RMM)

The information required is contained in this Safety Data Sheet.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**8.1. Control parameters****Exposure limits**List source(s): **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **IRE** - 2021 Code of Practice for the Chemical Agents Regulations, Schedule 1. Published by the Health and Safety Authority

Component	European Union	The United Kingdom	France	Belgium	Spain
Potassium Iodide					TWA / VLA-ED: 0.01 ppm (8 horas) TWA / VLA-ED: 0.1 mg/m ³ (8 horas)
Boric Acid				TWA: 2 mg/m ³ 8 uren STEL: 6 mg/m ³ 15 minuten	STEL / VLA-EC: 6 mg/m ³ (15 minutos). TWA / VLA-ED: 2 mg/m ³ (8 horas)
Iodine		STEL: 0.1 ppm 15 min STEL: 1.1 mg/m ³ 15 min	STEL / VLCT: 0.1 ppm. STEL / VLCT: 1 mg/m ³ .	TWA: 0.01 ppm 8 uren TWA: 0.1 mg/m ³ 8 uren STEL: 0.1 ppm 15 minuten STEL: 1 mg/m ³ 15 minuten	STEL / VLA-EC: 0.1 ppm (15 minutos). STEL / VLA-EC: 1 mg/m ³ (15 minutos). TWA / VLA-ED: 0.01 ppm (8 horas) TWA / VLA-ED: 0.1 mg/m ³ (8 horas)
Potassium Hydroxide		STEL: 2 mg/m ³ 15 min	STEL / VLCT: 2 mg/m ³ .		STEL / VLA-EC: 2 mg/m ³ (15 minutos).

Component	Italy	Germany	Portugal	The Netherlands	Finland
Potassium Iodide			TWA: 0.01 ppm 8 horas		
Boric Acid		TWA: 0.5 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 10 mg/m ³ (8 Stunden). MAK when boric acid and tetraborates are present together, the MAK value is 0.75 mg boron/m ³ Höhepunkt: 10 mg/m ³	STEL: 6 mg/m ³ 15 minutos TWA: 2 mg/m ³ 8 horas		
Iodine		Haut	STEL: 0.1 ppm 15 minutos TWA: 0.01 ppm 8 horas		STEL: 0.1 ppm 15 minuutteina STEL: 1.1 mg/m ³ 15 minuutteina Iho
Potassium Hydroxide			Ceiling: 2 mg/m ³		Ceiling: 2 mg/m ³

Component	Austria	Denmark	Switzerland	Poland	Norway
Boric Acid			STEL: 1.8 mg/m ³ 15 Minuten TWA: 1.8 mg/m ³ 8 Stunden		
Iodine	Haut MAK-KZGW: 0.1 ppm 15 Minuten MAK-KZGW: 1 mg/m ³ 15 Minuten MAK-TMW: 0.1 ppm 8 Stunden MAK-TMW: 1 mg/m ³ 8	Ceiling: 0.1 ppm Ceiling: 1 mg/m ³	Haut/Peau STEL: 0.1 ppm 15 Minuten STEL: 1 mg/m ³ 15 Minuten TWA: 0.1 ppm 8 Stunden TWA: 1 mg/m ³ 8	STEL: 1 mg/m ³ 15 minutach TWA: 0.5 mg/m ³ 8 godzinach	Ceiling: 0.1 ppm Ceiling: 1 mg/m ³

	Stunden Ceiling: 0.1 ppm Ceiling: 1 mg/m ³		Stunden		
Potassium Hydroxide	MAK-TMW: 2 mg/m ³ 8 Stunden	STEL: 2 mg/m ³ 15 minuter	TWA: 2 mg/m ³ 8 Stunden	STEL: 1 mg/m ³ 15 minutach TWA: 0.5 mg/m ³ 8 godzinach	Ceiling: 2 mg/m ³

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Potassium Iodide	TWA: 5.0 mg/m ³				
Boric Acid	TWA: 5.0 mg/m ³		TWA: 2 mg/m ³ 8 hr. STEL: 6 mg/m ³ 15 min		
Iodine	TWA: 3.0 mg/m ³	STEL-KGVI: 0.1 ppm 15 minutama. STEL-KGVI: 1.1 mg/m ³ 15 minutama.	TWA: 0.01 ppm 8 hr. inhalable fraction and vapour TWA: 0.01 mg/m ³ 8 hr. STEL: 0.1 ppm 15 min		TWA: 0.1 mg/m ³ 8 hodinách. Ceiling: 1 mg/m ³
Potassium Hydroxide	TWA: 2.0 mg/m ³	STEL-KGVI: 2 mg/m ³ 15 minutama.	STEL: 2 mg/m ³ 15 min		TWA: 1 mg/m ³ 8 hodinách. Ceiling: 2 mg/m ³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Iodine	STEL: 0.1 ppm 15 minutites. STEL: 1 mg/m ³ 15 minutites.		STEL: 0.1 ppm STEL: 1 mg/m ³ TWA: 0.1 ppm TWA: 1 mg/m ³	STEL: 1 mg/m ³ 15 percekben. CK TWA: 1 mg/m ³ 8 órában. AK lehetséges borön keresztüli felszívódás	STEL: 0.1 ppm STEL: 1 mg/m ³
Potassium Hydroxide	TWA: 2 mg/m ³ 8 tundides.		STEL: 2 mg/m ³ TWA: 2 mg/m ³	STEL: 2 mg/m ³ 15 percekben. CK TWA: 2 mg/m ³ 8 órában. AK	STEL: 2 mg/m ³

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Boric Acid	TWA: 10 mg/m ³	TWA: 10 mg/m ³ IPRD			
Iodine	TWA: 1 mg/m ³	Ceiling: 0.1 ppm Ceiling: 1 mg/m ³			TWA: 0.09 ppm 8 ore TWA: 0.5 mg/m ³ 8 ore STEL: 0.2 ppm 15 minute STEL: 1 mg/m ³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Potassium Iodide	MAC: 3 mg/m ³				
Boric Acid	MAC: 10 mg/m ³		TWA: 0.5 mg/m ³ 8 urah inhalable fraction STEL: 1.0 mg/m ³ 15 minutah inhalable fraction		
Iodine	Skin notation MAC: 1 mg/m ³	Ceiling: 1.1 mg/m ³ TWA: 0.1 ppm TWA: 1.1 mg/m ³		Binding STEL: 0.1 ppm 15 minuter Binding STEL: 1 mg/m ³ 15 minuter	
Potassium Hydroxide				Binding STEL: 2 mg/m ³ 15 minuter TLV: 1 mg/m ³ 8 timmar. NGV	

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

Derived No Effect Level (DNEL)

See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Boric Acid 10043-35-3 (0 - 10%)				DNEL = 392mg/kg bw/day
Iodine 7553-56-2 (0 - 10%)				DNEL = 0.01mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Boric Acid 10043-35-3 (0 - 10%)				DNEL = 8.3mg/m ³
Iodine 7553-56-2 (0 - 10%)				DNEL = 0.07mg/m ³
Potassium Hydroxide 1310-58-3 (0 - 10%)			DNEL = 1mg/m ³	

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Boric Acid 10043-35-3 (0 - 10%)	PNEC = 2.9mg/L		PNEC = 13.7mg/L	PNEC = 10mg/L	PNEC = 5.7mg/kg soil dw
Iodine 7553-56-2 (0 - 10%)	PNEC = 18.13µg/L	PNEC = 3.99mg/kg sediment dw		PNEC = 11mg/L	PNEC = 5.95mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water intermittent	Food chain	Air
Boric Acid 10043-35-3 (0 - 10%)	PNEC = 2.9mg/L				
Iodine 7553-56-2 (0 - 10%)	PNEC = 60.01µg/L	PNEC = 20.22mg/kg sediment dw			

8.2. Exposure controls

Engineering Measures None under normal use conditions

Personal protective equipment

Eye/face Protection Wear chemical splash goggles and face shield. If splashes are likely to occur.. Goggles.

Skin and body protection Wear protective gloves/protective clothing.

Respiratory Protection No protective equipment is needed under normal use conditions.
Recommended Filter type: Particle filter.

Environmental exposure controls No information available

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid
Appearance	Dark amber
Odor	None
Odor Threshold	No information available
pH	7.0
PH Range	5.5 - 8.5

Property	Values	Remarks • Method
Melting point/freezing point	No information available	
Boiling Point/Range	~ 100 °C / 212 °F	
Flash Point (High in °C)	No information available	
Evaporation Rate	No information available	
Flammability (solid, gas)	No information available	
Flammability Limit in Air		
Upper flammability limit:	No information available	
Lower flammability limit:	No information available	
Vapor pressure	No information available	
Vapor Density	No information available	
Specific Gravity	No information available	
Water Solubility	Soluble in water	
Solubility in other solvents	No information available	
Partition coefficient	No information available	
Autoignition Temperature	-	
Decomposition Temperature	No information available	
Kinematic viscosity	No information available	
Dynamic viscosity	No information available	
Explosive Properties	No information available	
Oxidizing Properties	No information available	

9.2. Other information

Softening Point	No information available
Molecular Weight	No information available
VOC Content(%)	No information available
Density	No Information available
Bulk Density	No information available

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

No information available

10.2. Chemical stability

Stable under normal conditions

Explosion Data

Sensitivity to Mechanical Impact	None
Sensitivity to Static Discharge	None

10.3. Possibility of hazardous reactions

None under normal processing

10.4. Conditions to avoid

Extremes of temperature and direct sunlight

10.5. Incompatible materials

No information available

10.6. Hazardous decomposition products

Thermal decomposition can lead to release of irritating gases and vapors

SECTION 11: TOXICOLOGICAL INFORMATION**11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008****Product Information**

Acute Toxicity No information available
Unknown Acute Toxicity 0 % of the mixture consists of ingredient(s) of unknown toxicity.
The following values are calculated based on chapter 3.1 of the GHS document
ATEmix (oral) 1,072.00 mg/kg

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water	LD50 > 90 mL/kg (Rat)		
Potassium Iodide		LD50 > 2000 mg/kg (Rat)	
Boric Acid	LD50 = 2660 mg/kg (Rat)	LD50 > 2000 mg/kg (Rabbit)	LC50 > 2.12 mg/L (Rat) 4 h
Iodine	LD50 = 14 g/kg (Rat)	LD50 > 2000 mg/kg (Rabbit) LD50 = 1425 mg/kg (Rabbit)	LC50 > 4.588 mg/L (Rat) 4 h
Potassium Hydroxide	LD50 = 284 mg/kg (Rat)		

Skin Corrosion/Irritation No information available
(b) skin corrosion/irritation; Category 2.
Serious eye damage/eye irritation No information available
Sensitization No information available
Mutagenic Effects No information available
Carcinogenic effects No information available
Reproductive Effects No information available
STOT - single exposure No information available
STOT - repeated exposure Causes damage to organs through prolonged or repeated exposure
Target Organs None known.
Aspiration hazard No information available

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12. ECOLOGICAL INFORMATION**12.1. Toxicity**

Ecotoxicity effects .

Harmful to aquatic life with long lasting effects Harmful to aquatic life
0% of the mixture consists of components(s) of unknown hazards to the aquatic environment

Component	Freshwater Algae	Freshwater Fish	Water Flea
Potassium Iodide	-	LC50: > 100 mg/L, 96h static (Danio rerio)	-
Boric Acid	-	LC50: = 1020 mg/L, 72h flow-through (Carassius auratus)	EC50: 115 - 153 mg/L, 48h (Daphnia magna)
Iodine	-	LC50: = 1.67 mg/L, 96h static (Oncorhynchus mykiss)	-
Potassium Hydroxide	-	LC50: = 80 mg/L, 96hr static (Gambusia affinis)	-

12.2. Persistence and degradability

12.3. Bioaccumulative potential

Component	log Pow	Bioconcentration factor (BCF)
Boric Acid	-1.09	0 dimensionless
Potassium Hydroxide	0.83	No data available

12.4. Mobility in soil

Mobility

Component	log Pow
Boric Acid 10043-35-3 (0 - 10%)	-1.09
Potassium Hydroxide 1310-58-3 (0 - 10%)	0.83

12.5. Results of PBT and vPvB assessment

No information available

12.6. Endocrine disrupting properties

This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant

This product does not contain any known or suspected substance

Ozone Depletion Potential

This product does not contain any known or suspected substance

SECTION 13. DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1 UN-No	Not Regulated
14.2 Proper Shipping Name	Not Regulated
14.3 Hazard Class	Not Regulated
14.4 Packing Group	Not Regulated
14.5 Marine Pollutant	Not Applicable
14.6 Special Provisions	None
14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code	No information available

ADR

14.1. UN number	Not Regulated
14.2. UN proper shipping name	Not Regulated
14.3. Transport hazard class(es)	Not Regulated
14.4. Packing group	Not Regulated

ICAO

14.1 UN-No	Not Regulated
14.2 Proper Shipping Name	Not Regulated
14.3 Hazard Class	Not Regulated
14.4 Packing Group	Not Regulated
14.5 Environmental hazard	Not Applicable
14.6 Special Provisions	None

IATA

14.1 UN-No	Not Regulated
14.2 Proper Shipping Name	Not Regulated
14.3 Hazard Class	Not Regulated
14.4 Packing Group	Not Regulated
14.5 Environmental hazard	Not Applicable
14.6 Special Provisions	None

SECTION 15: REGULATORY INFORMATION**15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture****International Inventories**

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS), U.S.A. (TSCA).

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Water	7732-18-5	231-791-2	-	-	X	X	KE-35400	X	-
Potassium Iodide	7681-11-0	231-659-4	-	-	X	X	KE-29149	X	X
Boric Acid	10043-35-3	233-139-2	-	-	X	X	KE-03499	X	X
Iodine	7553-56-2	231-442-4	-	-	X	X	KE-21023	X	-
Potassium Hydroxide	1310-58-3	215-181-3	-	-	X	X	KE-29139	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Water	7732-18-5	X	ACTIVE	X	-	X	X	X
Potassium Iodide	7681-11-0	X	ACTIVE	X	-	X	X	X
Boric Acid	10043-35-3	X	ACTIVE	X	-	X	X	X
Iodine	7553-56-2	X	ACTIVE	X	-	X	X	X
Potassium Hydroxide	1310-58-3	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

European Union

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Water	7732-18-5	-	-	-
Potassium Iodide	7681-11-0	-	-	-
Boric Acid	10043-35-3	-	Use restricted. See item 30. (see link for restriction details) Use restricted. See item 75. (see link for restriction details)	SVHC Candidate list - 233-139-2 - Toxic for reproduction, Article 57c
Iodine	7553-56-2	-	Use restricted. See item 75. (see link for restriction details)	-
Potassium Hydroxide	1310-58-3	-	Use restricted. See item 75. (see link for restriction details)	-

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work

Take note of Directive 94/33/EC on the protection of young people at work

Take note of Directive 94/33/EC on the protection of young people at work. Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work.

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification

Water endangering class = 3 (self classification)

Component	Germany - Water Classification (AwSV)
Potassium Iodide 7681-11-0 (40 - 50%)	WGK3
Boric Acid 10043-35-3 (0 - 10%)	WGK1
Iodine 7553-56-2 (0 - 10%)	WGK2

Potassium Hydroxide 1310-58-3 (0 - 10%)	WGK1
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Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Iodine 7553-56-2 (0 - 10%)	Prohibited and Restricted Substances		
Potassium Hydroxide 1310-58-3 (0 - 10%)	Prohibited and Restricted Substances		

15.2. Chemical safety assessment

A Chemical safety assessment according to regulation (EC) No. 1907/2006 is not required

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed
H315 - Causes skin irritation
H319 - Causes serious eye irritation
H360FD - May damage fertility. May damage the unborn child
H312 - Harmful in contact with skin
H314 - Causes severe skin burns and eye damage
H332 - Harmful if inhaled
H360Fd - May damage fertility. Suspected of damaging the unborn child
H372 - Causes damage to organs through prolonged or repeated exposure
H400 - Very toxic to aquatic life

Key or legend to abbreviations and acronyms used in the safety data sheet

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

TWA TWA (time-weighted average)

Ceiling Maximum limit value

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

STEL STEL (Short Term Exposure Limit)

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Full text of H-Statements referred to under section 3

H360FD - May damage fertility. May damage the unborn child

H302 - Harmful if swallowed

H314 - Causes severe skin burns and eye damage

H312 - Harmful in contact with skin

H332 - Harmful if inhaled

H400 - Very toxic to aquatic life

H372 - Causes damage to the kidneys/ liver/ eyes/ brain/ respiratory system/ central nervous system through prolonged or repeated exposure in contact with skin

H315 - Causes skin irritation

H319 - Causes serious eye irritation

Prepared By	Regulatory Affairs
Prepared For	Thermo Fisher Scientific Inc.
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Revision Date	07-Aug-2023
Reason for revision	SDS sections updated.
Training Advice	Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No
1907/2006 .**

Disclaimer

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End of Safety Data Sheet